$$= \frac{1}{2}\left\{(x_1y_2 + x_2y_3 + x_3y_4 + x_4y_1) - (x_2y_1 + x_3y_2 + x_4y_3 + x_1y_4)\right\}$$

$$= \frac{1}{2}\left\{(88 + 60 - 15 - 24) - (30 - 55 - 48 + 24)\right\}$$

$$= \frac{1}{2}\{109 + 49\}$$

$$= \frac{1}{2}\{158\} = 79 \text{ sq.units}$$

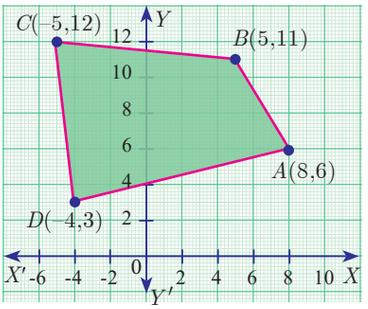


Fig. 5.12

Progress Check

Given a quadrilateral $ABCD$ with vertices $A(-3, -8)$, $B(6, -6)$, $C(4, 2)$, $D(-8, 2)$

1. Find the area of ΔABC.
2. Find the area of ΔACD.
3. Calculate area of ΔABC + area of ΔACD.
4. Find the area of quadrilateral $ABCD$.
5. Compare the answers obtained in 3 and 4.

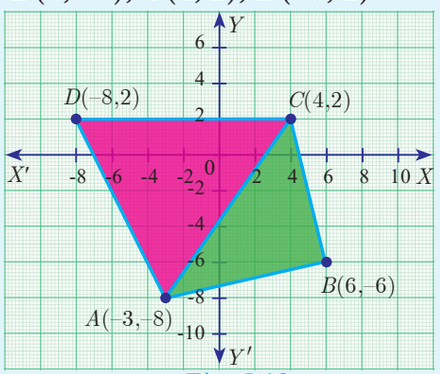


Fig. 5.13

Example 5.7 The given diagram shows a plan for constructing a new parking lot at a campus. It is estimated that such construction would cost ₹1300 per square feet. What will be the total cost for making the parking lot?

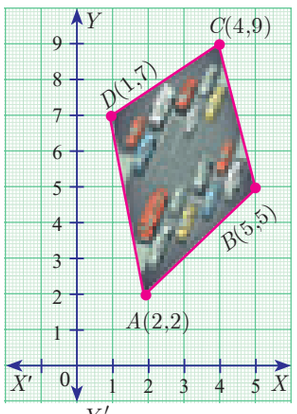


Fig. 5.14

Solution The parking lot is a quadrilateral whose vertices are at $A(2,2)$, $B(5,5)$, $C(4,9)$ and $D(1,7)$.

Area of parking lot $= \frac{1}{2}\begin{vmatrix} 2 & 5 & 4 & 1 & 2 \\ 2 & 5 & 9 & 7 & 2 \end{vmatrix}$ sq.units

$$= \frac{1}{2}\left\{(10 + 45 + 28 + 2) - (10 + 20 + 9 + 14)\right\}$$

$$= \frac{1}{2}\{85 - 53\}$$

$$= \frac{1}{2}(32) = 16 \text{ sq.units.}$$

Area of parking lot $= 16$ sq.feets

Construction rate per square feet $= ₹1300$

Total cost for constructing the parking lot $= 16 \times 1300 = ₹20800$

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